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Centre Number

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Student Number

2021

Year 12

Mathematics Extension 2

Task 2

18/3/21

Q	Marks
MC	/2
3	/14
4	/15
5	/14
6	/13
Total	/58

General Instructions:

- Reading time – 75 minutes
- Working time – 5 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- Answer all questions in the booklets provided

Total Marks:
58

Section I – 2 marks

- Allow about 3 minutes for this section

Section II – 58 marks

- Allow about 72 minutes for this section

This question paper must not be removed from the examination room.

This assessment task constitutes 20% of the course.

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Section I

2 marks

Allow about 3 minutes for this section

Use the multiple-choice sheet for Question 1–2

- 1 Consider the statement,

“There is no integer which is the largest integer”

Which of the following statements is equivalent to this statement?

(A) $\exists y: (\forall x, y > x)$

(B) $\forall x, (\exists y: y > x)$

(C) $\exists x: (\forall y, y > x)$

(D) $\forall y, (\exists x: y > x)$

- 2 If z_1 and z_2 are two non – zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then $\arg z_1 - \arg z_2$ equals

(A) 0

(B) $-\frac{\pi}{2}$

(C) $\frac{\pi}{2}$

(D) $-\pi$

End of Section I

Section II

In Questions 3 – 6, your response should include relevant mathematical reasoning and/or calculations.

Question 3: Answer Question 3 in a new booklet. (14 Marks)

(a) Let $z = 5 + i$ and $\omega = 3 + 2i$.

(i) Find $z - \bar{\omega}$ in $a + ib$ form. 2

(ii) Find $\frac{10\omega}{z - 2}$ in $a + ib$ form. 2

(iii) Find $\left| \frac{z}{\omega^8} \right|$ 2

(iv) Find $\arg(z\omega)$ 2

(v) Use the polar form of $z\omega$, and hence, find $|z\omega|^8$ 2

(b) Consider the complex number polynomial $p(z) = z^4 - 3z^3 + 6z^2 - 12z + 8$.

(i) Given that $p(2i) = 0$, factorise p into linear and quadratic factors with real coefficients. 3

(ii) Find all roots of p . 1

Question 4: Answer Question 4 in a new booklet. (15 Marks)

(a) $z^2 = -3 + 4i$.

(i) Express z in component form. 3

(ii) Hence, or otherwise, solve the quadratic equation 2
 $z^2 - 4z + (7 - 4i) = 0$

(b) (i) On an Argand diagram, shade the region where $|z - 1| \leq 1$ and $0 \leq \arg z \leq \frac{\pi}{6}$. 2

(ii) Find the perimeter of the shaded region 2

(c) Consider the factorisation

$$z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1)$$

(i) Show that the roots of the equation $z^6 + z^3 + 1 = 0$ are: 2

$$e^{\pm \frac{i2\pi}{9}}, e^{\pm \frac{i4\pi}{9}} \text{ and } e^{\pm \frac{i8\pi}{9}}.$$

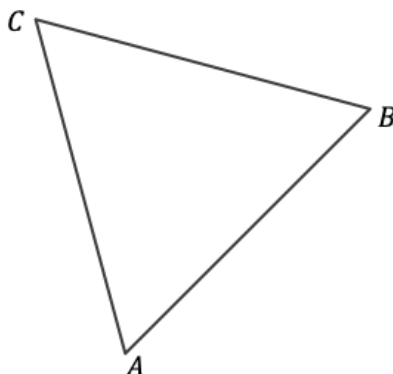
(ii) Divide $z^6 + z^3 + 1$ by z^3 and let $x = z + \frac{1}{z}$ 2
Find the cubic equation satisfied by x .

(iii) Deduce that 2

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$$

Question 5: Answer Question 5 in a new booklet. (14 Marks)

- (a) On the Argand diagram, the points A, B and C represent the complex numbers α, β and γ respectively. $\triangle ABC$ is equilateral, named with its vertices taken in anti-clockwise direction.



- (i) Show that

2

$$\gamma - \alpha = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) (\beta - \alpha).$$

- (ii) Show that

2

$$\alpha^2 + \beta^2 + \gamma^2 = \alpha\beta + \beta\gamma + \gamma\alpha$$

- (b) Consider the statement:

If n^3 is even, then n is even, $n \in \mathbb{Z}$

- (i) Write the contrapositive of this statement.

1

- (ii) Let $p: n^3$ is even, and $q: n$ is even

3

Prove the contrapositive.

- (iii) Hence, prove by contradiction that $\sqrt[3]{6}$ is irrational.

4

- (c) Every person on an island is either a knight or a knave. Knights always tell the truth, and knaves always lie. Alice and Bob are residents on the island.

2

Alice says: 'We are both knaves.'

Using method of Direct proof, determine what Alice and Bob are.

Question 6: Answer Question 6 in a new booklet. (13 Marks)

- (a) In the Tower of Hanoi puzzle, you have three pegs and a collection of n discs of different sizes. Initially the discs are stacked in size order on the left-hand peg.

- Discs can be moved one at a time from one peg to any other peg
- A larger disc can never rest on a smaller one.
- The third peg can be temporarily used.

The aim of the puzzle is to transfer all the discs to the second peg using the smallest possible number of moves.



Let a_n be the minimum number of moves needed to solve the Tower of Hanoi of n discs.

- (i) Prove that the number of moves needed for a Tower of Hanoi of $n + 1$ discs, a_{n+1} is given by 2

$$a_{n+1} = 2a_n + 1$$

- (ii) Use the equation $a_{n+1} = 2a_n + 1$, 3
to prove by mathematical induction that:

$$a_n = 2^n - 1, \forall n \in \mathbb{N}.$$

- (b) (i) If $a > 0$, $b > 0$ show that $\frac{a+b}{2} > \sqrt{ab}$. 2

- (ii) If $a > 0$, $b > 0$ and $c > 0$, show that $a^2 + b^2 + c^2 \geq ab + bc + ca$ 1

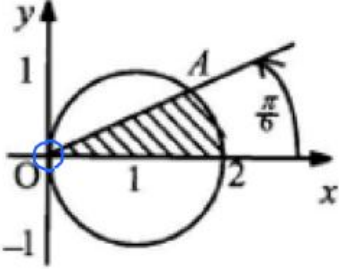
- (iii) Hence, show that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$. 2

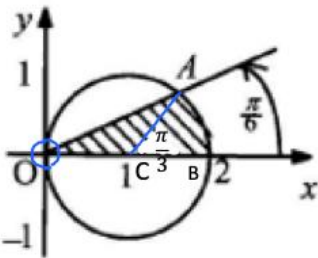
- (iv) Also, prove that $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a + b + c)$ 3

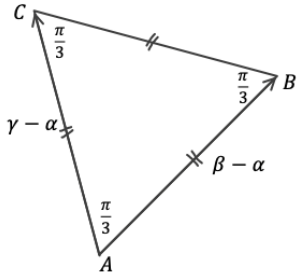
End of Examination

Year 12 Mathematics Extension 2 Marking Scheme Task 2 2021

Question		Marks	
1.	B		
2.	A Equality holds only when z_1 and z_2 are collinear, $z_2 = kz_1, k > 0$ Hence, $\arg z_1 - \arg z_2 = 0$		
3. (a) (i)	Let $z - \bar{\omega} = 5 + i - (3 - 2i) = 2 + 3i$		
(ii)	$\frac{10\omega}{z - 2} = \frac{10(3 + 2i)}{5 + i - 2}$ $= \frac{10(3 + 2i)}{3 + i}$ $= \frac{10(3 + 2i)(3 - i)}{10}$ $= (3 + 2i)(3 - i)$ $= 9 - 3i + 6i + 2$ $= 11 + 3i$	2 marks: correct solution with correct working 1 mark: rationalises the denominator, or a minor error	
(iii)	$\left \frac{z}{\omega^8} \right = \frac{ z }{ \omega ^8}$ $ z = \sqrt{5^2 + 1^2} = \sqrt{26}$ $ \omega = \sqrt{3^2 + 2^2} = \sqrt{13}$ $\frac{ z }{ \omega ^8} =$ $= \frac{\sqrt{5^2 + 1^2}}{(\sqrt{3^2 + 2^2})^8} = \frac{\sqrt{26}}{13^4}$	2 marks: correct answer from correct working 1 mark: calculates z and ω	
(iv)	$zw = (5 + i)(3 + 2i)$ $= 15 + 10i + 3i - 2$ $= 13 + 13i = 13(1 + i)$ $\arg(zw) = \tan^{-1} 1 = \frac{\pi}{4}$	1 mark: correctly calculates zw 1 mark: correctly calculates the argument	
(v)	$zw = \left(13, \frac{\pi}{4} \right)$ $zw = 13 \times 2 e^{\frac{i\pi}{4}}$ $(zw)^8 = 13^8 \left(\sqrt{2} e^{\frac{i\pi}{4}} \right)^8$ $= 13^8 \times 16 \times e^{2\pi i}$ $= 13^8 \times 16$	1 mark: expresses zw in polar form 1 mark: applies De-Moivre theorem and gives the correct answer	

(b) (i)	$p(z) = z^4 - 3z^3 + 6z^2 - 12z + 8.$ Since $z = 2i$ is a root of a polynomial with real coefficients, then $z = -2i$ is also a root. Hence, $(z - 2i)(z + 2i) = z^2 + 4$ is a factor. $z^4 - 3z^3 + 6z^2 - 12z + 8 = (z^2 + 4)(z^2 + az + 2)$ Comparing the coefficients of z, $-12 = 4a \rightarrow a = 3$ $z^2 + 3z + 2$ is a factor. $\therefore p(z) = (z^2 + 4)(z^2 + 3z + 2)$ $= (z^2 + 4)(z - 2)(z - 1)$	1 mark: states that $-2i$ is also a root with reason. 1 mark: states that $z^2 + 4$ is a factor 1 mark: correctly divides, or otherwise finds the other factor $(z^2 + 3z + 2)$ 1 mark: expresses $p(z)$ as factors with real coefficients	
(b) (ii)	$p(z) = 0$ Thus the four roots are $\pm 2i, 1, 2$	1 mark: finds the correct roots	
4. (a) (i)	$z^2 = -3 + 4i = (a + ib)^2$ $a^2 - b^2 = -3; 2ab = 4 \therefore ab = 2$ Hence, $a = 1, b = 2$ $-3 + 4i = (1 + 2i)^2$ $z = \sqrt{-3 + 4i}$ $= \sqrt{(1 + 2i)^2}$ $= \pm(1 + 2i)$	3 marks: correctly solves the equation and finds the square root. 2 mark: 1 mark:	
(ii)	$z^2 - 4z + (7 - 4i) = 0$ $z = \frac{4 \pm \sqrt{16 - 4(7 - 4i)}}{2}$ $z = 2 \pm \sqrt{4 - (7 - 4i)}$ $= 2 \pm \sqrt{-3 + 4i} \quad (1)$ $z = 2 \pm (1 + 2i)$ $z = 3 + 2i, 1 - 2i$	1 mark: applies the quadratic formula and gives (1) 1 mark: uses result from (a), to find z.	
(b) (i)		*draws the circle with centre (1,0) and radius 1 unit ** draws the line OA clearly marking $\frac{\pi}{6}$ as the angle subtended at the origin. *** shades the inside of the circle between OX and OA ***** Excludes the origin 2 marks: All aspects are correct 1 mark: at least two correct	

(ii)	 $\angle ACB = 2\angle AOB = \frac{\pi}{3}$ Hence $\angle AOC = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ $OA = \sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \times \cos \frac{2\pi}{3}} = \sqrt{3}$ Arc BA = $r\theta = 1 \times \frac{\pi}{3} = \frac{\pi}{3}$ Perimeter = $2 + \sqrt{3} + \frac{\pi}{3}$ units	1 mark: Calculates the length of chord OA showing all working with reasoning 1 mark: Calculates length of arc AB and calculates perimeter	
(c) (i)	$z^9 - 1 = 0$ $z^9 = 1 + 0i = e^{i2k\pi},$ where $k = 0, \pm 1, \pm 2, \pm 3, \pm 4$ Hence, $z = e^{\frac{i2k\pi}{9}}$, where $k = 0, \pm 1, \pm 2, \pm 3, \pm 4$ That is the 9 roots, starting from 1 are located on a unit circle at an angle of $\frac{2\pi}{9}$ between them. However, $z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1)$ Using the same argument, the three cube roots of unity (solutions of $z^3 - 1 = 0$ are $1, e^{\pm \frac{i2\pi}{3}} = e^{\pm \frac{i6\pi}{9}}$, that is when $k = \pm 3$ Hence, the remaining roots of $z^9 - 1 = 0$ are roots of $z^6 + z^3 + 1 = 0$ are $z = e^{\frac{i2k\pi}{9}}, \text{ where } k = \pm 1, \pm 2, \pm 4$ Hence, the roots are $e^{\pm \frac{i2\pi}{9}}, e^{\pm \frac{i4\pi}{9}}$ and $e^{\pm \frac{i8\pi}{9}}$.	1 mark: Finds the 9 roots of $z^9 - 1 = 0$ 1 mark: explains how the solutions of $z^3 - 1 = 0$ are got, and hence gives the roots of $z^6 + z^3 + 1 = 0$	
(ii)	Dividing $z^6 + z^3 + 1 = 0$ $z^3(z^3 + 1 + \frac{1}{z^3}) = 0$ $z \neq 0, \text{ hence } z^3 + 1 + \frac{1}{z^3} = 0$ That is, $z^3 + \frac{1}{z^3} = -1 \quad 1 \text{ mark}$ We have, $x^3 = \left(z + \frac{1}{z}\right)^3$ $x^3 = z^3 + \frac{1}{z^3} + 3z + \frac{3}{z}$ Hence, $x^3 = -1 + 3\left(z + \frac{1}{z}\right)$ $\therefore x^3 = -1 + 3x$	1 mark: divides $z^6 + z^3 + 1 = 0$ by z^3 and establishes $z^3 + \frac{1}{z^3} = -1$ (* must note $z \neq 0$) 1 mark: reorganises the equation and finds the polynomial that satisfies x	

	$x^3 - 3x + 1 = 0$ satisfies $x = z + \frac{1}{z}$		
(iii)	Let $z_1 = e^{\frac{i2\pi}{9}}, z_2 = e^{\frac{i4\pi}{9}}, z_3 = e^{\frac{i8\pi}{9}}$ Hence $z_1 + \frac{1}{z_1} = 2 \cos \frac{2\pi}{9}$, $z_2 + \frac{1}{z_2} = 2 \cos \frac{4\pi}{9}, z_3 + \frac{1}{z_3} = 2 \cos \frac{8\pi}{9}$ These are the three roots of the polynomial $x^3 - 3x + 1 = 0$. Sum of roots, $2 \left(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} \right) = 0$ Hence, $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$ However, $\cos \frac{8\pi}{9} = \cos \left(\pi - \frac{\pi}{9} \right) = -\cos \frac{\pi}{9}$ $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} - \cos \frac{\pi}{9} = 0$ $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$	2 marks: Correct answer from correct working 1 mark: shows that $z + \frac{1}{z} = 2 \cos \frac{2k\pi}{9}$ and attempts to find sum of roots	
5. (a) (i)	 $\triangle ABC$ is equilateral, hence $AC = AB$ and the angle between these sides is $\frac{\pi}{3}$. $\therefore \overrightarrow{AC}$ is the rotation of $\overrightarrow{AB}$ in the anticlockwise direction by $\frac{\pi}{3}$. $\gamma - \alpha = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) (\beta - \alpha).$	1 mark: relates the complex numbers $\gamma - \alpha$ and $\beta - \alpha$ to the vectors $\overrightarrow{AC}$ and $\overrightarrow{AB}$ 1 mark: explains properties of equilateral triangle and vector relationships to deduce the result	
(ii)	Similarly, $\overrightarrow{CB}$ is the rotation of $\overrightarrow{CA}$ in the anticlockwise direction by $\frac{\pi}{3}$. $\therefore \beta - \gamma = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) (\alpha - \gamma).$ Hence, $\frac{\gamma - \alpha}{\beta - \alpha} = \frac{\beta - \gamma}{\alpha - \gamma} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ (using result from (a)) $(\gamma - \alpha)(\alpha - \gamma) = (\beta - \alpha)(\beta - \gamma)$ $-\alpha^2 - \gamma^2 + 2\gamma\alpha = \beta^2 - \alpha\beta + \beta\gamma + \gamma\alpha$ $\alpha^2 + \beta^2 + \gamma^2 = \alpha\beta\gamma + \beta\gamma + \gamma\alpha$	1 mark: writes a similar result to (a), considering a second pair of triangle sides as vectors 1 mark: equates the two results to obtain the required result	
(b) (i)	If n^3 is even, then n is even, $n \in \mathbb{Z}$ Contrapositive: *	1 mark: for either statement	

	If n is not even, then n^3 is not even. Or If n is odd, then n^3 is odd.	the statement cannot begin with 'if n^3 is even)	
(ii)	p: n^3 is even, hence, $\neg p$: n^3 is odd and q: n is even, hence, $\neg q$: n is odd ** n is odd, $n = 2k + 1, k \in \mathbb{Z}$ Then, $n^3 = (2k + 1)^3$ $= 8k^3 + 12k^2 + 6k + 1$ $= 2(4k^3 + 6k^2 + 3k) + 1$ $= 2M + 1, M \in \mathbb{Z}. ***$ Hence, n^3 is odd, $\neg q$ is true. ie. $\neg q \Rightarrow \neg p$ *** The contrapositive is true, hence, the Proposition $p \Rightarrow q$ is true.	1 mark: uses mathematical notations and correctly expresses n as an odd number. 1 mark: proves n^3 is odd, by expressing all statements using correct notations. 1 mark: clearly states the equivalence, the proposition and its contrapositive.	
(iii)	Proof by contradiction: Let p: $\sqrt[3]{6}$ is irrational We need to prove that $\neg p$ is false. $\neg p$: $\sqrt[3]{6}$ is rational Assume $\neg p$ is true. * $\therefore \sqrt[3]{6} = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $p \neq kq, k \in \mathbb{Z}$ (p and q have no common factors) $\sqrt[3]{6} = \frac{p}{q}$ $p^3 = 6q^3$ (1) ** $\Rightarrow p^3$ is divisible by 2 $\Rightarrow p$ is divisible by 2 (using the result (b)) $\Rightarrow p = 2k, k \in \mathbb{Z}$ $\Rightarrow (2k)^3 = 6q^3$ (substituting in (1)) $\Rightarrow 8k^3 = 6q^3$ $\Rightarrow 4k^3 = 3q^3$ $\Rightarrow q^3$ is divisible by 2 $\Rightarrow q$ is divisible by 2 (using result (b)) So, p and q are both divisible by 2, which contradicts the fact that p and q have no common factors.	1 mark: writes the $\neg p$: statement as assumes it is true. 1 mark: establishes $p^3 = 6q^3$ giving reasons and with correct notations 2 marks: Correctly proves the contradiction 1 mark: Minor error in the contradiction proof	

(c)	Case1: Suppose Alice is a knight. $\Rightarrow$ Alice is telling the truth. $\Rightarrow$ Alice and Bob are both knaves. $\Rightarrow$ Alice is a knave and a knight. This is impossible. Case 2: Suppose Alice is a knave $\Rightarrow$ Alice is not telling the truth. $\Rightarrow$ Alice and Bob are not both knaves. $\Rightarrow$ Bob is a knight.		
6. (a) (i)	Suppose there are $n+1$ discs on the left-hand peg. If we want to be able to move the largest disc to the right-hand peg, then first we must transfer the other n discs to the centre peg. This takes a minimum of a_n moves. It takes 1 move to transfer the largest disc to the right-hand peg. Now we can complete the puzzle by transferring the n discs on the centre peg to the right-hand peg. This takes a minimum of a_n moves. Hence the minimum number of moves required to transfer all the discs is $a_{n+1} = a_n + a_n + 1$ $= 2a_n + 1$	1 mark: clearly explains the process and proves the result 1 mark: some explanation is given to prove the result	
(ii)	For each natural number n, let $P(n)$ be the proposition: $a_n = 2^n - 1$ Step 1: The minimum number of moves required to solve the Tower of Hanoi puzzle with one disc is 1. Since $2^1 - 1 = 1$, it follows that $P(1)$ is true. * Step 2: Let k be any natural number, and assume $P(k)$ is true. That is, $a_k = 2^k - 1 \text{ **}$ We now wish to prove that $P(k+1)$ is true, that is, $a_{k+1} = 2^{k+1} - 1$ We have LHS of $P(k+1) = a_{k+1}$ $= 2a_k + 1 \text{ ***}$	1 mark: verifies the result for $n=1$ 2 mark: writes the assumption statement, proves the induction and gives the conclusion. (ALL STEPS) Award 1 mark: correct assumption, attempts to prove for $n = k+1$ and gives the conclusion. (*, **, ***, *****)	

	$= 2(2^k - 1) + 1 \text{ using } P(k)$ $= 2^{k+1} - 2 + 1 \text{ ****}$ $= 2^{k+1} - 1$ $= RHS \text{ of } P(k + 1)$ We have proved that if $P(k)$ is true, then $P(k + 1)$ is true, $\forall n \in \mathbb{N}$ Conclusion: By the principle of mathematical induction, it follows that $P(n)$ is true for all $n \in \mathbb{N}$. Hence, we have shown that $a_n = 2^n - 1$, $\forall n \in \mathbb{N}$. *****		
(b) (i)	It is clear that $(a + b)^2 = (a - b)^2 + 4ab \Rightarrow (a + b)^2 > 4ab$ (equality iff $a = b$), since $(a - b)^2 \geq 0$. Then, if $a > 0$, $b > 0$, we get $\frac{a+b}{2} > \sqrt{ab} \text{ (equality iff } a = b \text{)}.$		
(ii)	It is clear that $a^2 + b^2 \geq 2ab$, $b^2 + c^2 \geq 2bc$, $a^2 + c^2 \geq 2ac$. By addition $2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$. Hence $a^2 + b^2 + c^2 \geq ab + bc + ca$ with equality iff $a = b = c$.		
(iii)	$a^3 + b^3 + c^3 - 3abc$ $= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ Since $a + b + c$ is positive, and $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$, the right-hand side of identity is not negative. Hence $a^3 + b^3 + c^3 \geq 3abc$ (equality iff $a = b = c$). The substitution $a^3 \rightarrow a$, $b^3 \rightarrow b$, $c^3 \rightarrow c$ gives $a + b + c \geq 3a^{1/3}b^{1/3}c^{1/3}$. Hence $\frac{a+b+c}{3} \geq \sqrt[3]{abc} \text{ (equality iff } a = b = c \text{)}.$		
(iv)	From (b), $ab + bc + ca \leq a^2 + b^2 + c^2$ (equality iff $a = b = c$). Exchanging in (2) a by a^2, b by b^2 and c by c^2 yields $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2$. 1 mark Also, from (b), we have		

	$ab \leq \frac{a^2+b^2}{2}, \quad bc \leq \frac{b^2+c^2}{2}, \quad ca \leq \frac{c^2+a^2}{2}. \quad 1$ mark Hence, $abc(a+b+c) = a^2bc + b^2ac + c^2ab \leq a^2\left(\frac{b^2+c^2}{2}\right) + b^2\left(\frac{a^2+c^2}{2}\right) + c^2\left(\frac{a^2+b^2}{2}\right) =$ $a^2b^2 + b^2c^2 + a^2c^2 \text{ (equality iff } a = b = c \text{)}.$ Hence $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a+b+c)$. 1 mark		
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